

Simple Model of a Helical Flux Compression Generator

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Consider a solenoid with N turns of wire per unit length, a total length X , and a cross sectional area A . If a current I runs through the wire, the interior magnetic field is

$$B = \mu_0 N I , \quad (1)$$

where $\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$ is the permeability of free space. The total magnetic flux through NX current loops is

$$\Phi = NXBA = \mu_0 N^2 AXI . \quad (2)$$

The self-inductance L is defined by $\Phi = LI$, and for the solenoid we have $L = \mu_0 N^2 AX$.

The detonation wave moves down the length of the solenoid with speed D . As the wave propagates so does the contact circle between the solenoid wire and the conductive armature surrounding the HE. In effect, the length of the solenoid is changing such that

$$X(t) = X_0 - Dt = X_0(1 - t/\tau) , \quad (3)$$

where $\tau = X_0/D$ is the burn time of the HE. It follows that

$$L = L_0(1 - t/\tau) . \quad (4)$$

The change in magnetic flux that accompanies the change in L produces an electric field along the length of the wire that opposes this change. The induced voltage is

$$V_{\text{ind}} = -d\Phi/dt . \quad (5)$$

If the power supply that produces the seed current is turned off at the time the HE is detonated, then the total voltage in the circuit is

$$V_{\text{tot}} = -d\Phi/dt = IR , \quad (6)$$

where R is the resistance of the wire. For simplicity, we will neglect the dependence of R on I and assume that R is proportional to the length of wire, so that

$$R = R_0(1 - t/\tau) . \quad (7)$$

Equation (6) provides us with a differential equation for the current:

$$\dot{I} + I \left(\frac{\dot{L}}{L} + \frac{R}{L} \right) = 0 . \quad (8)$$

Substituting the relations for L and R , we find

$$\frac{\dot{I}}{I} = \frac{1}{\tau} \left(1 - \frac{t}{\tau} \right)^{-1} - \frac{R_0}{L_0} . \quad (9)$$

The solution is

$$I(t) = I_0 \left(1 - \frac{t}{\tau} \right)^{-1} e^{-R_0 t/L_0} . \quad (10)$$